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Have you generated or been educated?

Teaching students how to use genAI to grow and to learn

P. Lafeber & H. vd Geest | ICERI 2026

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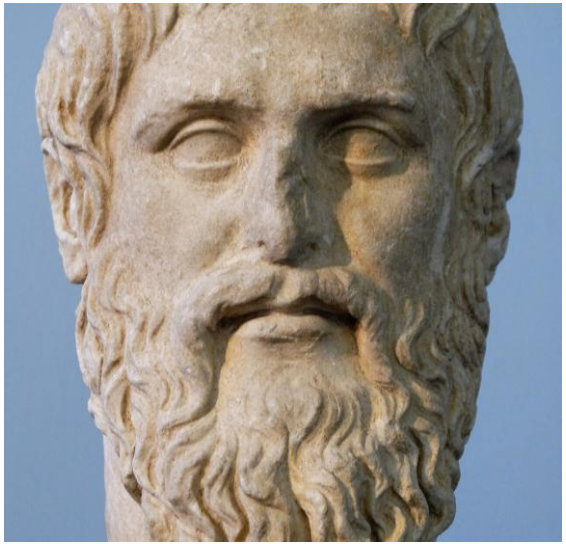
Welcome to the world of AI

A study coach in HvA AI Chat

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History of tools

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Writing will make us forget.

(Plato)



Too many books will confuse us.

(Conrad Gessner, 1545)

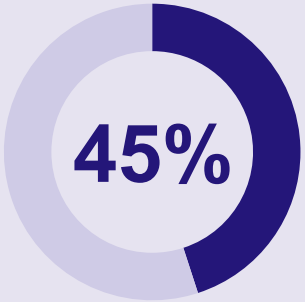


No more mental arithmetic.

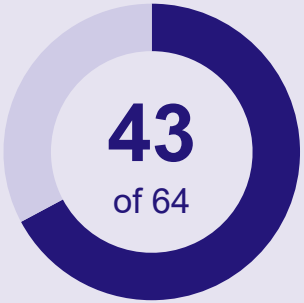
(HP-35 calculator, 1972)

Our students already use it

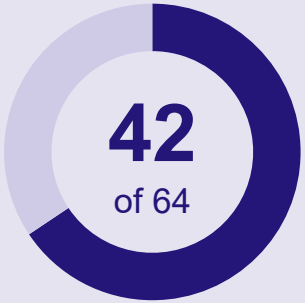
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use GenAI for their studies weekly or daily



found the chatbot too affirming



said a safe HvA tool made no difference to how they use AI

First survey, 64 first-year students

Why study?

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General AI chat

Student

Make a profile page in HTML and CSS for me.

AI

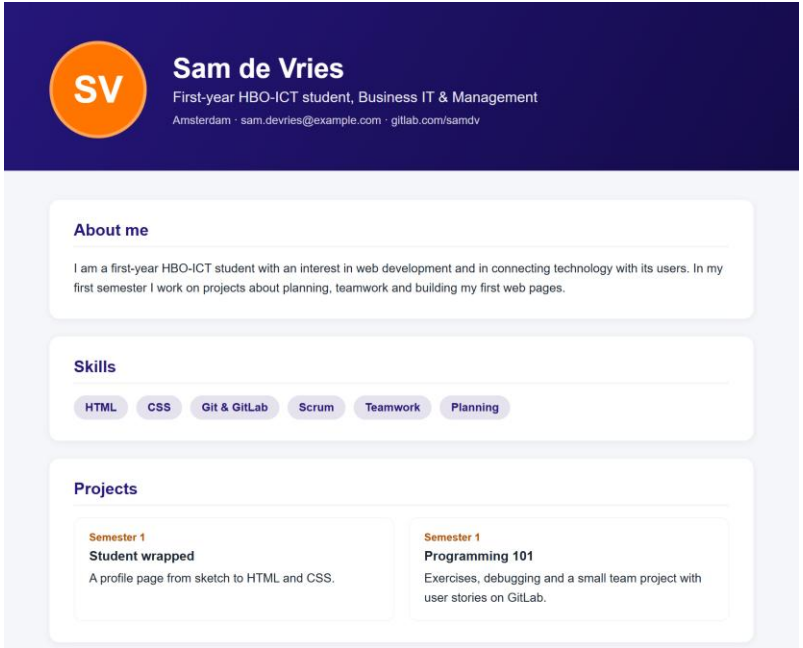
Sure! Here is a complete profile page you can use right away:

```
<header>
  <h1>Sam de Vries</h1>
  <p>First-year HBO-ICT student</p>
</header>
<section id="about"> ... </section>
<section id="projects"> ... </section>

header  { background: #25167A; color: #fff; }
section { padding: 24px; border-radius: 12px; }
/* ... and the rest of the styling ... */
```

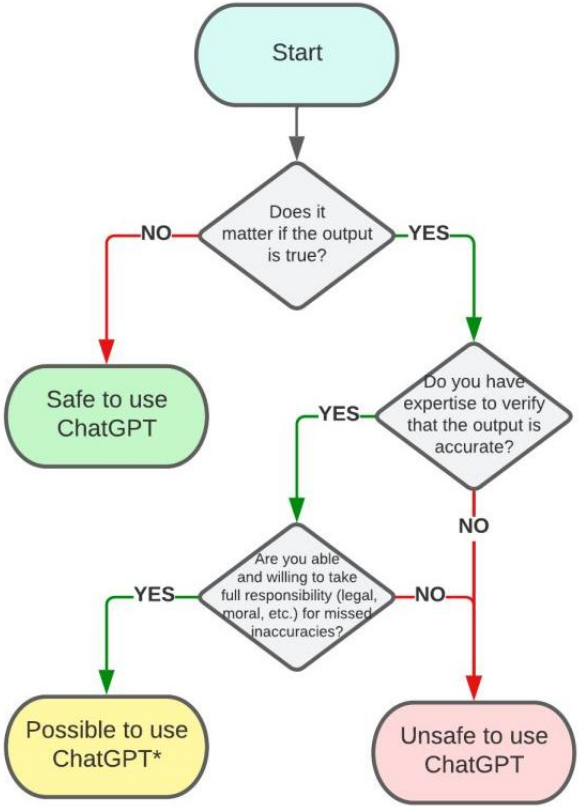
Copy both files into your project and you are done.





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Consider this first



Flowchart: Aleksandr Tiulkanov, CC BY

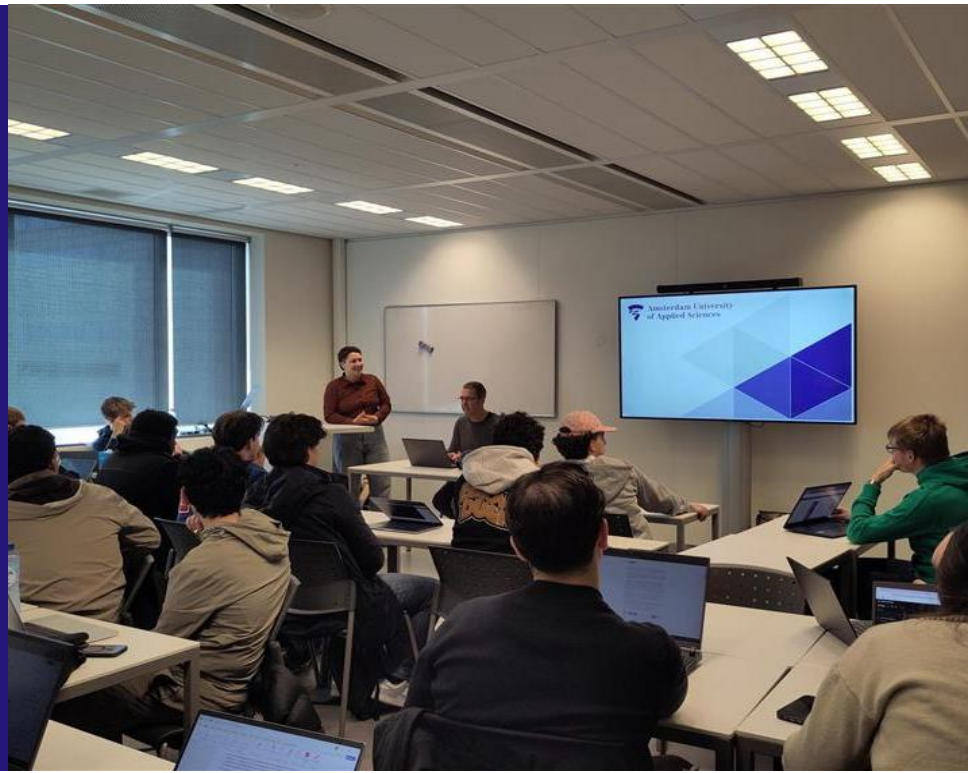
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Three classes, one question

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BIM, Cyber Security and Fusion, 2025-2026



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HvA AI Chat

+ Nieuwe Persona

Zoeken

Mijn persona's

- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- HvA Data Impact & Rapportage Skill v2
- SEO- en GEO-redacteur hva.nl (voorbeel...
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- HvA Analyse-assistent
- HvA Analyse-assistent
- HvA SEO & GEO Opleidingsreviewer
- HvA Opleidingspagina SEO/GEO-auditor
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Nieuwe Persona
- Studiecoach v4
- Studiecoach v3
- Scriptie
- StudieCoach

Gedeelde persona's

- HvA Data Impact & Rapportage Skill
- SEO-strateeg 2.0
- SEO-strateeg - opleidingen hva.nl
- AI-redacteur - opleidingen hva.nl
- Default fallback persona

Studiecoach v4

Korte werkinstructie voor de gebruiker

Chat starten

Mijn chats met deze persona

BPMN voor ING-inloggen bespreken

Eerdere maker-chats

Nog geen maker-sessies voor deze persona.

Stel je vraag

gpt-5.1

Persona eigenaar

h.s.a.van.der.geest

Taalmodel

gpt-5.1

Laatste update

02-07-2026

Beschikbaarheid persona

N/A

Gekoppelde groepen

Geen gekoppelde groepen

Bestanden

Geen geüploade bestanden

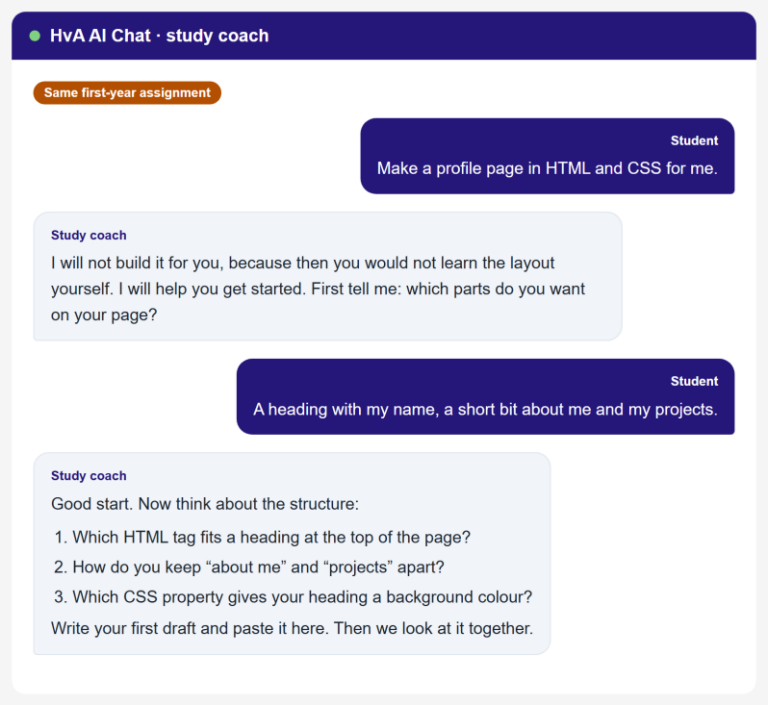
Links

Geen links

Bekijk persona

A coach that asks questions

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GROW

G — **Goal**
What do you want to learn?

R — **Reality**
What can you do already?

O — **Options**
Which ways can you try?

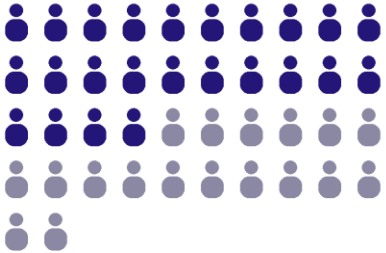
W — **Way forward**
What is your next step?

Coach versus open use

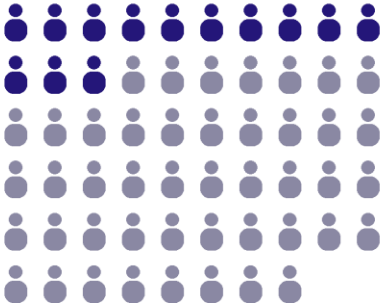
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Baseline test → **Intervention** → **Final test**

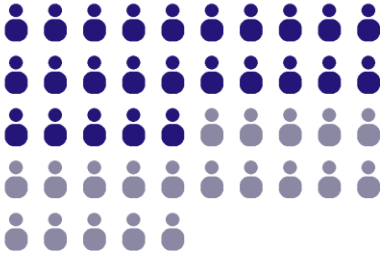
BIM
24 A + 18 B



Cyber Security
13 A + 45 B



Fusion
25 A + 20 B



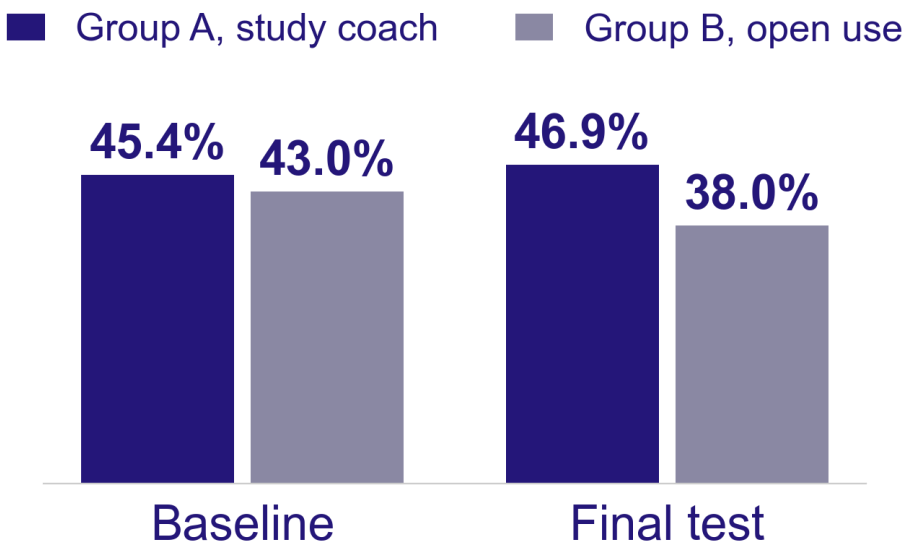
145
students

A: study coach
B: open use

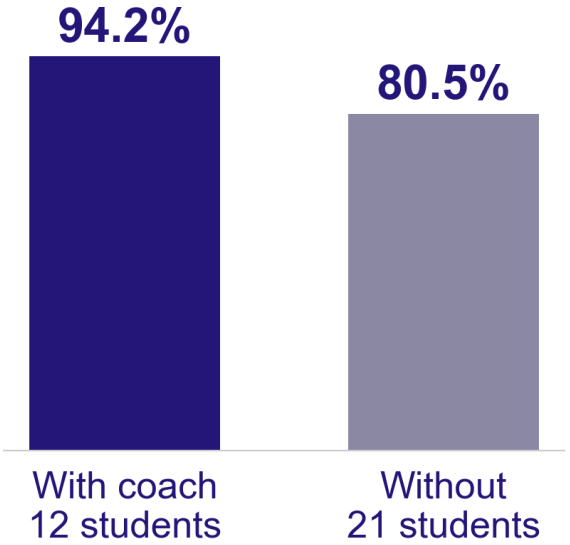
Scores held or went up

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Fusion, UML



Cyber Security, statistics



The coach was used

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72% of group A used the coach

15% of group B

Fusion, feedback session

With the coach we saw more

- **reflection**
- **check-ins**
- **intermediate steps**

But for code, it still gave full solutions.

A first signal, not proof

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- Existing classes, no random assignment
- Small groups
- A different test per programme



Four conditions for success

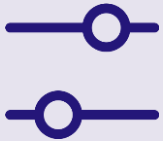
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01 Clear role



The prompt gives the AI the role of coach. Tell students why.

02 Per task



Coach works for planning and reflection. For code, set stricter rules.

03 No full answers



Hints and check questions instead of the full solution.

04 Show your steps



Prompts and steps are part of what students hand in.

Paper and teacher guide

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Includes the coach prompt

Scan for the paper and the guide

hugovandergeest.github.io/iceri2026



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Questions?

When students show their steps, the thinking stays with them.

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Paper and teacher guide

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